Operating Systems
Week 13
Lecture/Discussion 13

Junaid Abdul Wahid

Previous

- Simulating Thrashing, Memory Mayhem

Today Agenda

- Individual Assignment: Understanding Volatile and Non-Volatile Memory

Assignment Objective

To explore, differentiate, and critically analyze the two fundamental types of computer memory — **volatile** and **non-volatile** — and understand their practical applications in real-world computing systems.

Assignment Tasks

Each student is required to prepare a structured report addressing the following sections:

•Clearly define the following in your own words:

- **Volatile Memory**
- **Non-Volatile Memory**

•Explain *why* volatile memory loses its contents after power loss, and *why* non-volatile memory does not.

Assignment Tasks

Prepare a comparison table with **at least 6 technical and practical aspects:**

Feature	Volatile Memory	Non-Volatile Memory
Data Retention		
Speed		
Cost per GB		
Common Use Cases		
Power Dependency		
Typical Devices		

Assignment Tasks

Real-Life Examples

List **2 examples** of volatile memory.
• List **2 examples** of non-volatile memory.
•
•For each example:

- Mention **where it is used** (laptop, mobile, embedded device, etc.)
- Explain **its role** in the functioning of the system.